TOPOLOGY OF THE FIBRE OVER A POINT IN TERMS OF THE RESIDUAL SPACE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Residual space definition. definition-residual-space (same chapter, structural). The closed subspace t({x}).
- Topology of presentations. Properties of Spaces, lemma topology-points (cross-chapter, named). Specializations in the presentation topology.

Source reference. Original lemma 03JV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 732–732. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01K1: focused-seeds/geometry/S9-decent-topology-fibre/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06RV: focused-seeds/geometry/S9-decent-topology-fibre/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IR: focused-seeds/geometry/S9-decent-topology-fibre/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-topology-fibre/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $y \in |Y|$ and assume that y is represented by a quasi-compact monomorphism $\mathrm{Spec}(k) \to Y$. Then $|X_k| \to |X|$ is a homeomorphism onto $f^{-1}(\{y\}) \subset |X|$ with induced topology.*

Proof. We will use Properties of Spaces, Lemma 03IR and Morphisms of Spaces, Lemma 06RV without further mention. Let $V \to Y$ be an étale morphism with V affine such that there exists a $v \in V$ mapping to y. Since $\mathrm{Spec}(k) \to Y$ is quasi-compact there are a finite number of points of V mapping to y (Lemma 03JV). After shrinking V we may assume v is the only one. Choose a scheme U and a surjective étale morphism $U \to X$. Consider the commutative diagram

$$\begin{array}{ccccc} U & \longleftarrow & U_V & \longleftarrow & U_v \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & X_V & \longleftarrow & X_v \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & V & \longleftarrow & v \end{array}$$

Since $U_v \to U_V$ identifies U_v with a subset of U_V with the induced topology (Schemes, Lemma 01K1), and since $|U_V| \to |X_V|$ and $|U_v| \to |X_v|$ are surjective and open, we see that $|X_v| \to |X_V|$ is a homeomorphism onto its image (with induced topology). On the other hand, the inverse image of $f^{-1}(\{y\})$ under the open map $|X_V| \to |X|$ is equal to $|X_v|$. We conclude that $|X_v| \to f^{-1}(\{y\})$ is open. The morphism $X_v \to X$ factors through X_k and $|X_k| \to |X|$ is injective with image $f^{-1}(\{y\})$ by Properties of Spaces, Lemma 03H4. Using $|X_v| \to |X_k| \to f^{-1}(\{y\})$ the lemma follows because $X_v \to X_k$ is surjective. □